SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Solution Requirements

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

Functional and Non-Functional Requirements:

- **Functional Requirements (FR)** Define what the system should do.
- **Non-Functional Requirements (NFR)** Define how the system should perform.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	User Input	The system shall allow users to enter text describing their emotions, feelings, or learning difficulties.	High
2	Emotion Detection	The system shall analyze the input text using a fine-tuned BERT model and identify the user's emotion with a confidence score.	High
3	External Interfaces	The system shall integrate with the Gemini API to provide AI-powered learning support and guidance.	High

4	Recommendation Processing	The system shall generate personalized learning recommendations based on the detected emotion.	High
5	Reporting	The system shall display the detected emotion, confidence score, recommendations, and AI-generated response to the user.	Medium
6	Business Rules	The system shall recommend appropriate learning resources according to the detected emotional state.	High
7	Data Privacy	The system shall securely process user input and API communication without exposing sensitive information.	Medium
8	User Interface	The system shall provide a simple, interactive, and user-friendly interface for emotion analysis and learning support.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria

1	Performance & Speed	The system shall process user input and generate emotion detection results efficiently.	Response time ≤ **30 seconds**
2	Scalability	The system shall support multiple users without significant performance degradation.	Supports up to **100 concurrent users**
3	Security & Data Privacy	The system shall protect user data and ensure secure communication with external APIs.	Secure API key management and protected user data
4	Reliability & Availability	The system shall provide stable emotion detection and learning support with minimal failures.	**99% system availability** during operation
5	Usability & Accessibility	The system shall provide an intuitive and responsive interface accessible through modern web browsers.	Users can complete emotion analysis in **less than 2 minutes**
6	Maintainability	The system shall use modular, well-documented code that is easy to maintain and extend.	Modular architecture compatible with Windows and Linux